

NAME: Shruthe

ROLL NO: 241801265

FACE Prep

Operating System - Coding | REC | 3 Feb

Course Progress

Get Back

Back To Practice

Q1

Save

First-Come,First-Served CPU Scheduling Algorithm

Write a C program to implement the First-Come-First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
 $TAT = CT - AT$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
 $WT = TAT - BT$

Select language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5
6     printf("Enter number of processes: ");
7     scanf("%d", &n);
8
9     int AT[n], BT[n], CT[n], TAT[n], WT[n];
10     float avgTAT = 0, avgWT = 0;
11
12     for(i = 0; i < n; i++) {
13         printf("\nEnter Arrival Time and Burst Time for P%d: ", i+1);
14         scanf("%d %d", &AT[i], &BT[i]);
15     }
16
17     if(AT[0] == 0)
18         CT[0] = BT[0];
19     else
20         CT[0] = AT[0] + BT[0];
21
22     for(i = 1; i < n; i++) {
23         if(CT[i-1] < AT[i])
24             CT[i] = AT[i] + BT[i];
```

Select Language: C (GCC 9.2.0)

```
21
22     for(i = 1; i < n; i++) {
23         if(CT[i-1] < AT[i])
24             CT[i] = AT[i] + BT[i];
25         else
26             CT[i] = CT[i-1] + BT[i];
27     }
28
29     for(i = 0; i < n; i++) {
30         TAT[i] = CT[i] - AT[i];
31         WT[i] = TAT[i] - BT[i];
32
33         avgTAT += TAT[i];
34         avgWT += WT[i];
35     }
36
37     avgTAT /= n;
38     avgWT /= n;
39
40     printf("\n\nAverage Turnaround Time: %.2f\n", avgTAT);
41     printf("Average Waiting Time: %.2f\n", avgWT);
42
43     return 0;
44 }
```